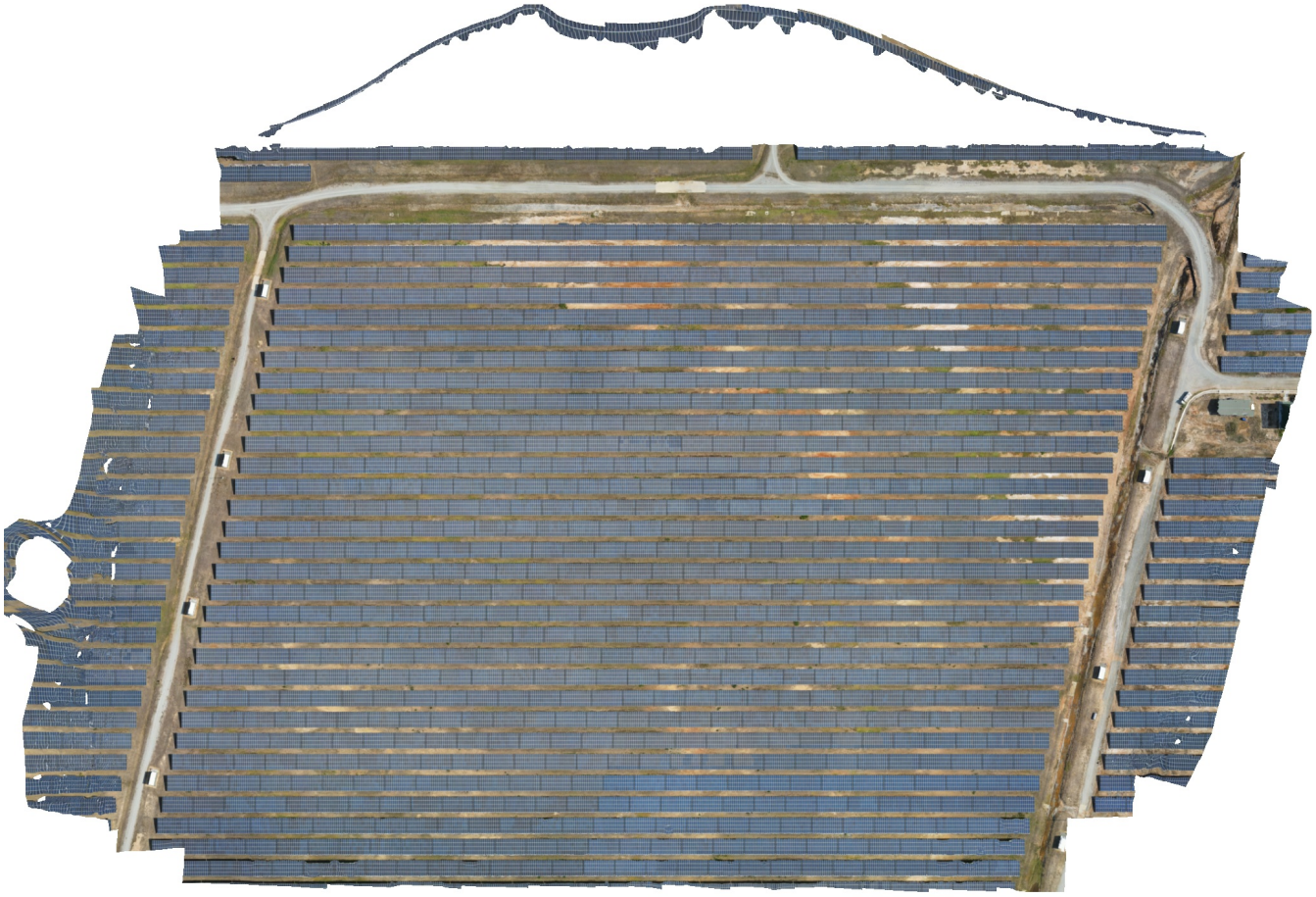
# Agisoft Metashape

**Processing Report**

**08 September 2026**



# Survey Data

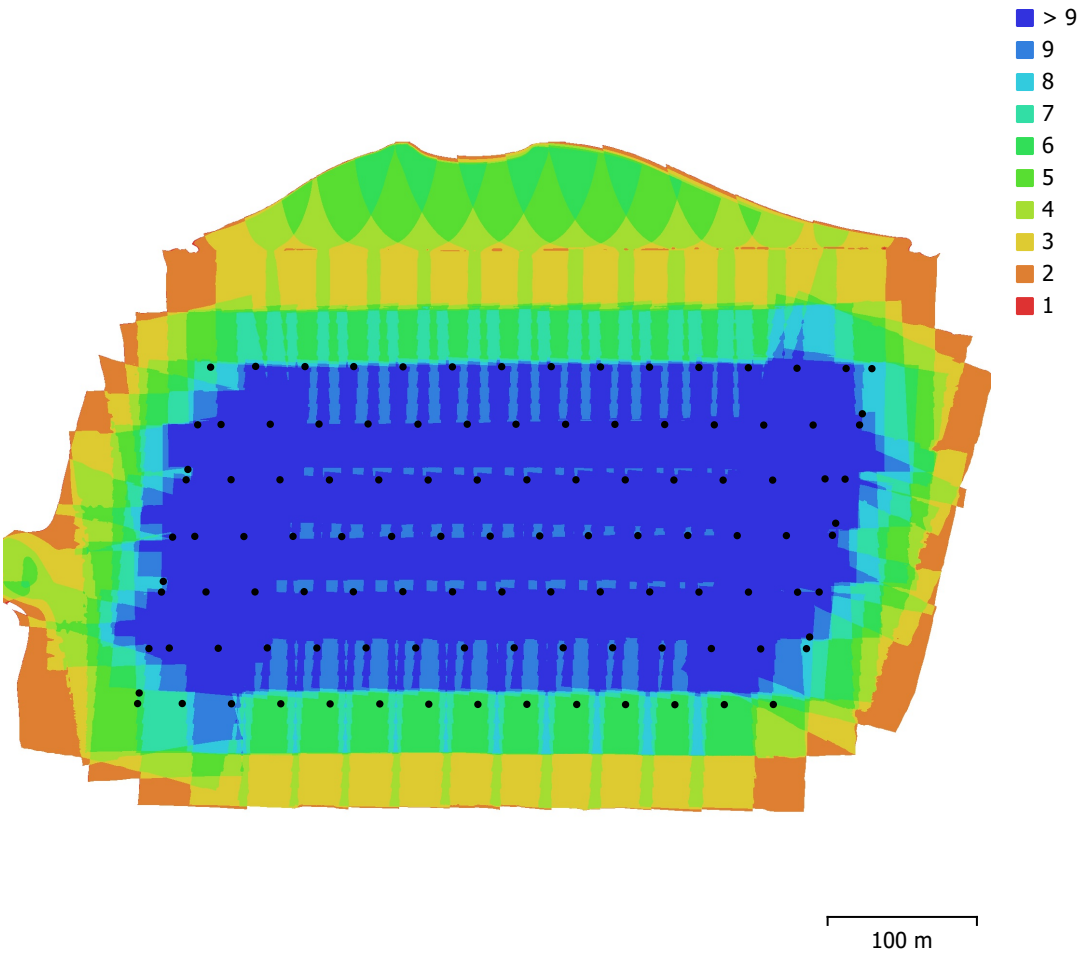



Fig. 1. Camera locations and image overlap.

| | | | |
|---|---|---|---|
| Number of images: | 110 | Camera stations: | 110 |
| Flying altitude: | 105 m | Tie points: | 63,566 |
| Ground resolution: | 1.80 cm/pix | Projections: | 164,032 |
| Coverage area: | 0.237 km² | Reprojection error: | 0.61 pix |

| Camera Model | Resolution | Focal Length | Pixel Size | Precalibrated |
|---|---|---|---|---|
| MAVIC2-ENTERPRISE-A... | 8000 x 6000 | 4.5 mm | 0.811 x 0.811 µm | No |

Table 1. Cameras.

# Camera Calibration

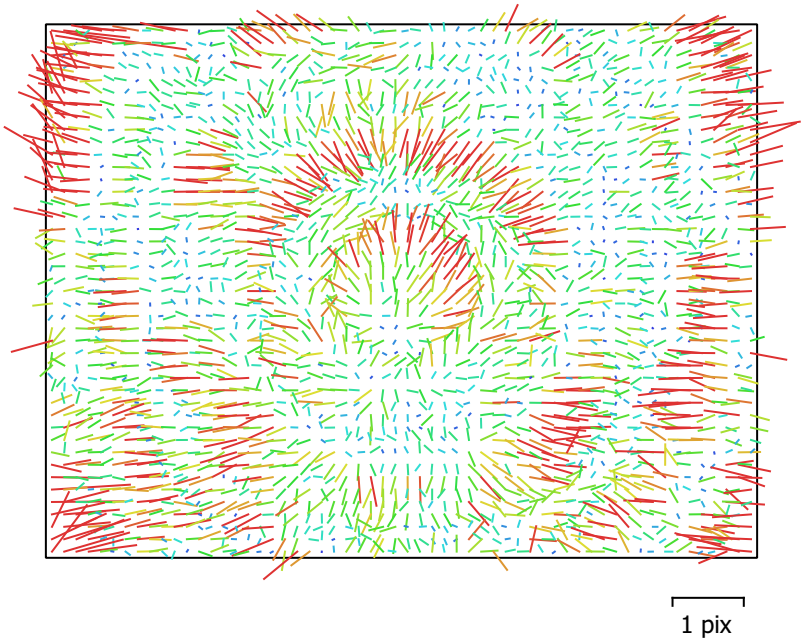



Fig. 2. Image residuals for MAVIC2-ENTERPRISE-ADVANCED (4.5mm).

**MAVIC2-ENTERPRISE-ADVANCED (4.5mm)**

110 images

| Type | Resolution | Focal Length | Pixel Size |
|---|---|---|---|
| **Frame** | **8000 x 6000** | **4.5 mm** | **0.811 x 0.811 µm** |

| | Value | Error | Cx | Cy | B1 | B2 | K1 | K2 | P1 | P2 |
|---|---|---|---|---|---|---|---|---|---|---|
| **F** | **5547** | | | | | | | | | |
| **Cx** | **-30.9558** | 0.76 | 1.00 | 0.07 | 0.14 | -0.42 | -0.05 | -0.04 | -0.31 | -0.08 |
| **Cy** | **48.3096** | 0.56 | | 1.00 | 0.28 | 0.04 | 0.01 | 0.01 | -0.09 | -0.14 |
| **B1** | **-4.5775** | 0.044 | | | 1.00 | -0.02 | -0.04 | -0.02 | -0.03 | -0.15 |
| **B2** | **2.41434** | 0.039 | | | | 1.00 | 0.02 | 0.01 | 0.13 | 0.06 |
| **K1** | **-0.00814363** | 0.00017 | | | | | 1.00 | -0.63 | -0.27 | 0.51 |
| **K2** | **-0.013165** | 3.9e-05 | | | | | | 1.00 | 0.11 | -0.26 |
| **P1** | **-0.000145752** | 5.1e-06 | | | | | | | 1.00 | -0.09 |
| **P2** | **-0.000360316** | 5.2e-06 | | | | | | | | 1.00 |

Table 2. Calibration coefficients and correlation matrix.

# Camera Locations

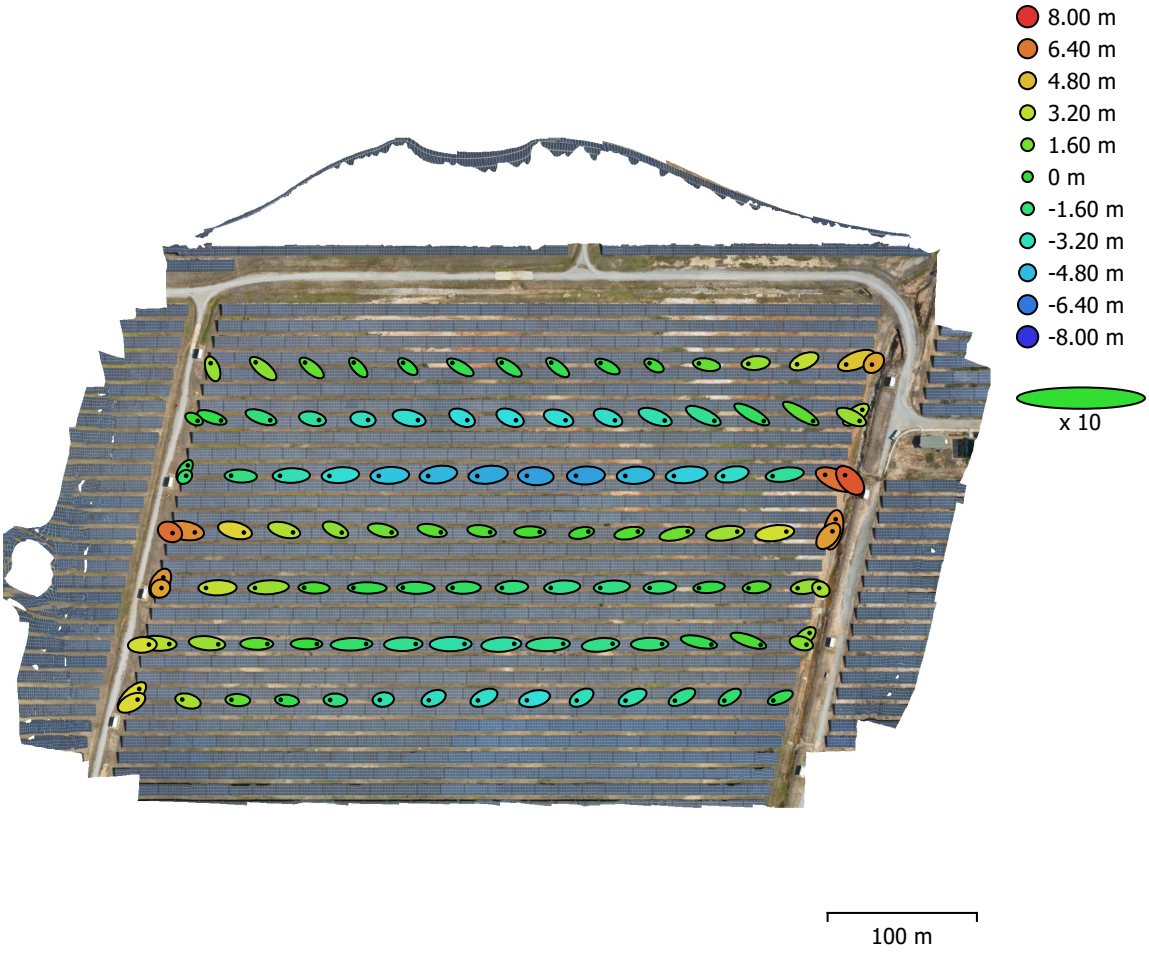



Fig. 3. Camera locations and error estimates.
Z error is represented by ellipse color. X,Y errors are represented by ellipse shape.
Estimated camera locations are marked with a black dot.

| X error (m) | Y error (m) | Z error (m) | XY error (m) | Total error (m) |
|---|---|---|---|---|
| 1.19483 | 0.397513 | 3.03443 | 1.25922 | 3.28534 |

Table 3. Average camera location error.
X - Longitude, Y - Latitude, Z - Altitude.

# Digital Elevation Model

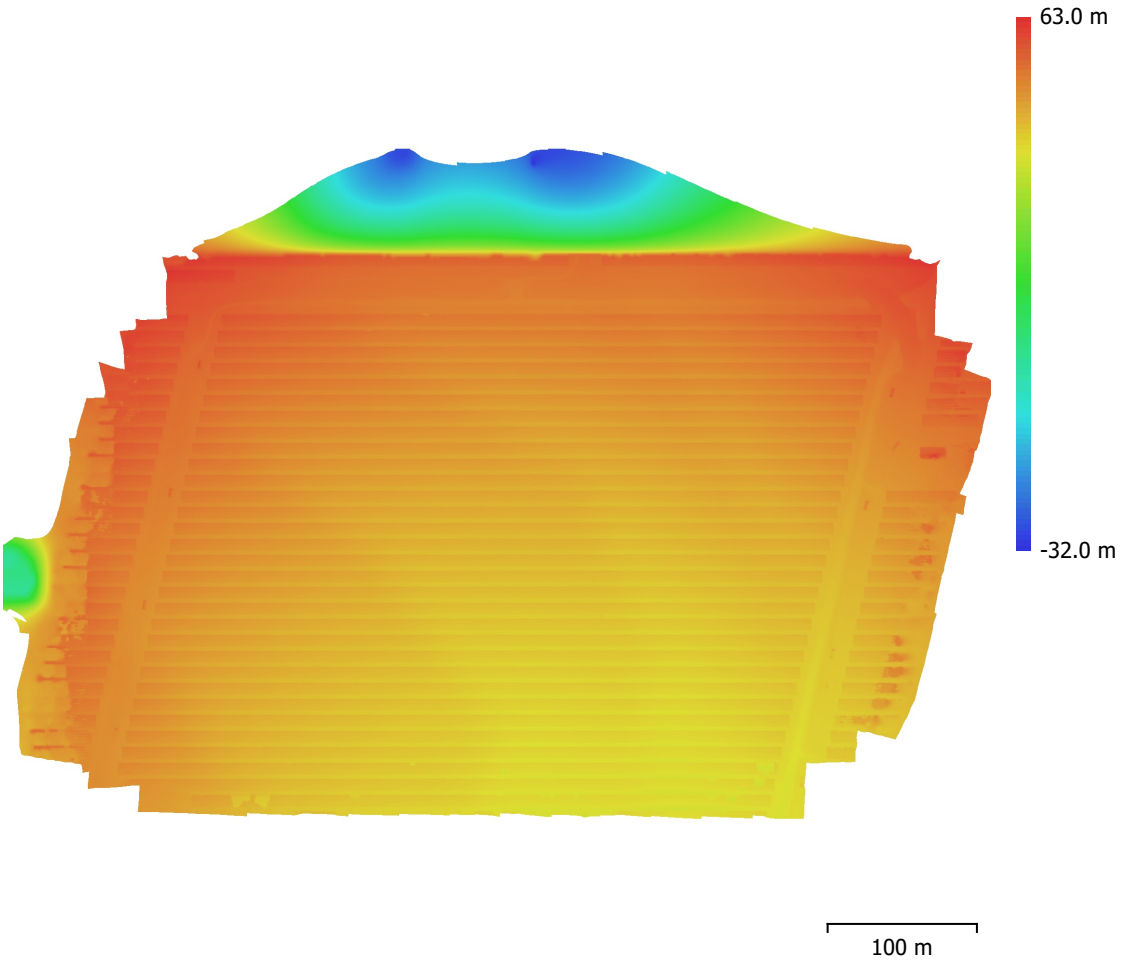



Fig. 4. Reconstructed digital elevation model.

| | |
|---|---|
| Resolution: | 3.80 cm/pix |
| Point density: | 693 points/m² |

# Processing Parameters

| | |
|---|---|
| **General** | |
| Images | 110 |
| Aligned images | 110 |
| Coordinate system | WGS 84 (EPSG::4326) |
| Rotation angles | Yaw, Pitch, Roll |
| **Tie Points** | |
| Points | 63,566 of 139,780 |
| RMS reprojection error | 0.213 (0.609868 pix) |
| Max reprojection error | 0.942987 (3.62725 pix) |
| Mean key point size | 2.9433 pix |
| Point colors | 3 bands, uint8 |
| Key points | 361.71 MB |
| Average tie point multiplicity | 2.64383 |
| **Alignment parameters** | |
| Accuracy | High |
| Generic preselection | Yes |
| Reference preselection | Source |
| Key point limit | 40,000 |
| Key point limit per Mpx | 1,000 |
| Tie point limit | 4,000 |
| Exclude stationary tie points | Yes |
| Guided image matching | No |
| Adaptive camera model fitting | Yes |
| Matching time | 3 minutes 33 seconds |
| Matching memory usage | 378.19 MB |
| Alignment time | 45 seconds |
| Alignment memory usage | 103.36 MB |
| **Optimization parameters** | |
| Parameters | f, cx, cy, k1, k2, p1, p2 |
| Adaptive camera model fitting | Yes |
| Exclude corners | No |
| Optimization time | 1 seconds |
| Date created | 2026:09:05 08:50:55 |
| Software version | 2.3.1.22580 |
| File size | 370.63 MB |
| **Depth Maps** | |
| Count | 110 |
| **Depth maps generation parameters** | |
| Quality | High |
| Filtering mode | Aggressive |
| Max neighbors | 16 |
| Processing time | 18 minutes 51 seconds |
| Memory usage | 6.35 GB |
| Date created | 2026:09:05 23:34:42 |
| Software version | 2.3.1.22580 |
| File size | 1.87 GB |
| **Point Cloud** | |
| Points | 181,840,813 |
| Coordinate precision | 4.61 m |
| **Point attributes** | |

| | |
|---|---|
| Color | 3 bands, uint8 |
| Normal | |
| **Point classes** | |
| Created (never classified) | 181,840,813 |
| **Depth maps generation parameters** | |
| Quality | High |
| Filtering mode | Aggressive |
| Max neighbors | 16 |
| Processing time | 18 minutes 51 seconds |
| Memory usage | 6.35 GB |
| **Point cloud generation parameters** | |
| Source data | Depth maps |
| Processing time | 22 minutes 41 seconds |
| Memory usage | 9.39 GB |
| Date created | 2026:09:06 00:28:11 |
| Software version | 2.3.1.22580 |
| File size | 2.35 GB |
| **DEM** | |
| Size | 17,400 x 11,796 |
| Resolution | 3.80 cm/pix |
| Coordinate system | WGS 84 (EPSG::4326) |
| **Depth maps generation parameters** | |
| Quality | High |
| Filtering mode | Aggressive |
| Max neighbors | 16 |
| Processing time | 18 minutes 51 seconds |
| Memory usage | 6.35 GB |
| **Reconstruction parameters** | |
| Source data | Depth maps |
| Interpolation | Enabled |
| Processing time | 54 minutes 22 seconds |
| Memory usage | 7.64 GB |
| Date created | 2026:09:06 01:22:48 |
| Software version | 2.3.1.22580 |
| File size | 697.66 MB |
| **Orthomosaic** | |
| Size | 34,800 x 23,592 |
| Resolution | 1.90 cm/pix |
| Coordinate system | WGS 84 (EPSG::4326) |
| Colors | 3 bands, uint8 |
| Orthophotos | 1.59 GB |
| **Reconstruction parameters** | |
| Blending mode | Mosaic |
| Surface | DEM |
| Enable hole filling | Yes |
| Enable ghosting filter | No |
| Processing time | 7 minutes 1 seconds |
| Memory usage | 4.49 GB |
| Date created | 2026:09:06 01:29:42 |
| Software version | 2.3.1.22580 |
| File size | 2.18 GB |
| **System** | |
| Software name | Agisoft Metashape Professional |
| Software version | 2.3.0 build 21954 |
| OS | Windows 64 bit |
| RAM | 15.78 GB |

| | |
|---|---|
| CPU | 11th Gen Intel(R) Core(TM) i5-11400H @ 2.70GHz |
| GPU(s) | NVIDIA GeForce RTX 3050 Laptop GPU |